

Detection of Suspicious Activities in Public Spaces Using Video Surveillance

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Abstract. Automatic detection of suspicious human activities in public spaces is crucial for enhancing security and safety. While many modern systems rely on deep learning, this project proposes a traditional computer vision approach using background subtraction, optical flow, and trajectory-based motion analysis to detect suspicious activities in surveillance videos. By focusing on handcrafted features and classical object tracking techniques, the system aims to provide a computationally efficient, interpretable, and real-time solution for identifying potentially risky human behaviors in public spaces.

Keywords: Video Surveillance · Action Recognition · Object Tracking · Suspicious Activity Detection · Classical Computer Vision, Motion Analysis, Optical Flow, Background Subtraction

1 Problem Description

Detecting suspicious human activities in public spaces is a critical task for improving security. Most current systems rely on deep learning models that require large labeled datasets and substantial computational resources.

This project explores a traditional computer vision approach that avoids deep learning. The focus is on detecting human motion patterns and classifying activities based on handcrafted features such as trajectories, optical flow, and spatio-temporal motion patterns. The objective is to develop an efficient system capable of identifying abnormal or suspicious behaviors, such as fighting, running, or loitering, in surveillance videos.

2 Literature Review

The paper [2] shows that it is very important that we prepare the model for the real environment where we don't include restrictions like only one person in the video and most importantly we don't want to use the same camera quality for training the model as in the real world it is very much likely to be affected by weather conditions, technical problems and so on. Temporal templates are used to classifying the moving objects, they use Motion History Image and Motion

Energy Image, which shows where the motion occurred recent will have higher pixel value for the Motion History Image than other static parts they yield Recurrent Motion Images.

[3] Covers all the hand features overview summary.

2.1 Problems Summaries

Background subtraction cannot deal with, Uninteresting moving Object, changes in the lighting condition and shadows.

Since Occlusion Occurs when object is not visible in the image, this is a difficult task.

Gaussian background subtraction cannot deal with movements of uninteresting objects.

3 Background

Traditional computer vision approaches for activity detection rely on motion analysis and object tracking rather than end-to-end neural networks.

Motion Detection: Background subtraction methods have been widely used for detecting moving objects in video sequences. Adaptive Gaussian Mixture Models (GMM) and running average methods can separate foreground objects from static backgrounds, which is essential for detecting human motion [5].

Optical Flow: Optical flow is a method of estimating apparent motion of scene points from a sequence of Images (Frames). Motion field $\rightarrow$ Optical flow constrain $\text{equ} \rightarrow$ lucas-kanade method $\rightarrow$ Coarse fine of Optical flow Optical flow algorithms, such as the Lucas-Kanade method, estimate pixel-wise motion between consecutive frames. Flow vectors can be used to understand movement patterns, helping distinguish between normal and suspicious activities.

Trajectory-Based Activity Recognition: By tracking moving objects and extracting trajectories, one can model human behavior over time. Deviations from normal trajectories can indicate abnormal or suspicious activities [7].

Object Tracking: Classical object tracking methods, such as Kalman Filters, Mean-Shift, or CamShift, are essential for maintaining consistent object identities across frames, enabling analysis of temporal motion patterns.

Existing Datasets: Authentically Distorted Surveillance Videos (AD-SVD) dataset [1] can be adapted for traditional methods, allowing evaluation of motion-based techniques under real-world video distortions.

4 Background

Differential Luminance Field Trajectory (DLFT) and Luminance Aligned Projection Distance (LAPD) methodologies [8] were each video clip is a trajectory in a high dimensional space for extracting information for action recognition.

The work of action recognition [4].

The work of Human Activity recognition used for detection of suspicious activities, We want to be able to detect whether a person is suspicious or Not. Being a Suspicious person can be defined where we detect if the people are fighting, ... , These activities are defined based on the Dataset I will be using for the project. Suspicious Human Activities recognition involves identification of human activities that can be classified as Normal or Abnormal (0,1) [ESTAR-555].

Having an accurate motion detection is needed; The paper proposes a background subtraction method that adapts the detection threshold by modeling both foreground and background probabilities and enforcing spatial consistency using Markov random fields, leading to more accurate motion detection [5].

Detecting suspicious activities from surveillance cameras is a very useful task and has received significant attention in computer vision. Current surveillance systems rely mostly on human operators monitoring multiple cameras, which makes it easy to miss events that may appear risky. Takai [6] proposed a system that first extracts motion regions corresponding to moving people, then performs behavior analysis, and finally estimates a risk score for each activity to prioritize potentially dangerous situations.

Object Tracking Methods [7] are very crucial when building a detecting suspicious activities.

This covers the abstract for the paper : QUALITY AWARE Features for performance prediction and time reduction [1] The existing body of work on video object tracking (VOT) algorithms has studied various image conditions such as occlusion, clutter, and object shape, which influence video quality and affect tracking performance. Nonetheless, there is no clear distinction between the performance reduction caused by scene-dependent challenges such as occlusion and clutter, and the effect of authentic in-capture and post-capture distortions. Despite the plethora of VOT methods in the literature, there is a lack of detailed studies analyzing the performance of videos with authentic in-capture and post-capture distortions.

We introduced a new dataset of authentically distorted videos (AD-SVD) to address this issue. This dataset contains 4476 videos with different authentic distortions and surveillance activities. Furthermore, it provides benchmarking results for evaluating ten state-of-the-art visual object trackers (from VOT 2017–2018 challenges) based on the proposed dataset. In addition, this study develops an approach for performance prediction and quality-aware feature selection for single-object tracking in authentically distorted surveillance videos. The method predicts the performance of a VOT algorithm with high accuracy. Then, the probability of obtaining the reference output is maximized without executing the tracking algorithms. We also propose a framework to reduce video tracker computation resources (time and video storage space). We achieve this by balancing processing time and tracking accuracy by predicting the perfor-

mance in a range of spatial resolutions. This approach can reduce the execution time by up to 34

5 Proposed Solution

The proposed system will consist of the following steps:

1. **Preprocessing:** Stabilize the video and reduce noise using standard filters.
2. **Foreground Detection:** Apply background subtraction methods (e.g., Gaussian Mixture Model) to extract moving objects.
3. **Object Tracking:** Track detected moving objects using classical tracking methods such as Kalman Filter or Mean-Shift to generate trajectories.
4. **Feature Extraction:** Compute motion features such as optical flow, speed, direction, and trajectory shape.
5. **Activity Classification:** Use rule-based or statistical models to classify activities as normal or suspicious. Abnormal behavior patterns can include sudden accelerations, unusual movement directions, or interactions with other objects.

This approach avoids the need for deep neural networks while remaining computationally efficient and interpretable.

6 Justification

Traditional computer vision methods offer several advantages:

- **Computational Efficiency:** No need for GPUs or large datasets for training.
- **Interpretability:** The system's decisions are based on clearly defined motion patterns and rules.
- **Real-Time Feasibility:** Classical techniques can run efficiently on low-power devices, suitable for real-time surveillance.
- **Adaptability:** Methods such as adaptive background subtraction can handle changing lighting and camera conditions.

While deep learning offers high accuracy, the traditional approach provides a lightweight, understandable, and robust alternative for real-time suspicious activity detection.

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